PROJECT DEFINITION

Hot Hand, Real or Believed?

A behavioural analysis of setting decisions in professional volleyball

Studies on Human Behaviour · Project exam

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MOTIVATION

Why volleyball is interesting for studying behaviour

Volleyball provides an interesting setting for studying behaviour because performance and decision-making can be observed separately.

During each rally, Data Volley records the sequence of technical actions and their outcomes. At the same time, the setter repeatedly decides which attacker receives the next ball.

This makes it possible to ask not only whether recent success predicts future performance, but also whether it influences the setter’s next decision.

Performance → decision → performance

PREVIOUS ATTACK

↓

OUTCOME

Success / no success

↓

SETTER’S NEXT DECISION

Who receives the ball?

↓

NEXT ATTACK

The hot hand hypothesis

> *“A successful outcome increases the probability of another success shortly afterwards.”*

In statistical terms, this means asking whether consecutive outcomes are independent or whether recent performance helps predict the next outcome.

In volleyball, the question has an additional behavioural component: after observing an attacker’s performance, the setter decides who will receive the next set.

QUESTIONS

Does success persist?

Does the setter behave as if it does?

Where the question comes from

1985

Gilovich, Vallone & Tversky

Basketball shooting data show no clear relationship between consecutive shots. The hot hand is interpreted as a cognitive illusion: people tend to see streaks even in random sequences

2012

Raab, Gula & Gigerenzer

The question is studied in volleyball, where defensive adjustments are more limited than in basketball. The authors find streaks for about half of the elite players analysed and show that playmakers use recent performance when allocating the ball

2012 – 2014

Köppen & Raab
MacMahon, Köppen & Raab

Video-based experiments show that players tend to allocate more balls to a «hot» attacker, even when base rates are kept equal. The hot-hand belief can also change with framing and experience

THE GAP THIS PROJECT ADDRESSES

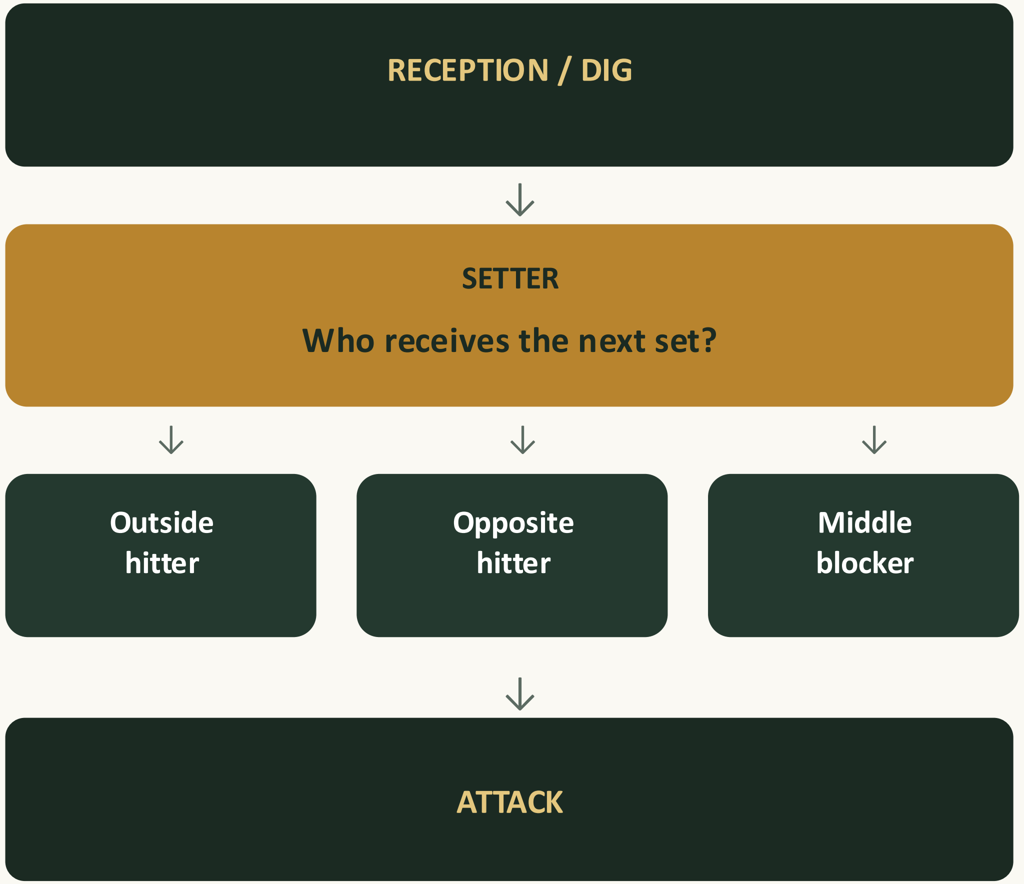
Most evidence on how the hot-hand belief affects allocation decisions comes from controlled experiments. This project studies the same question using a full season of real setting decisions

The setter as a decision-maker

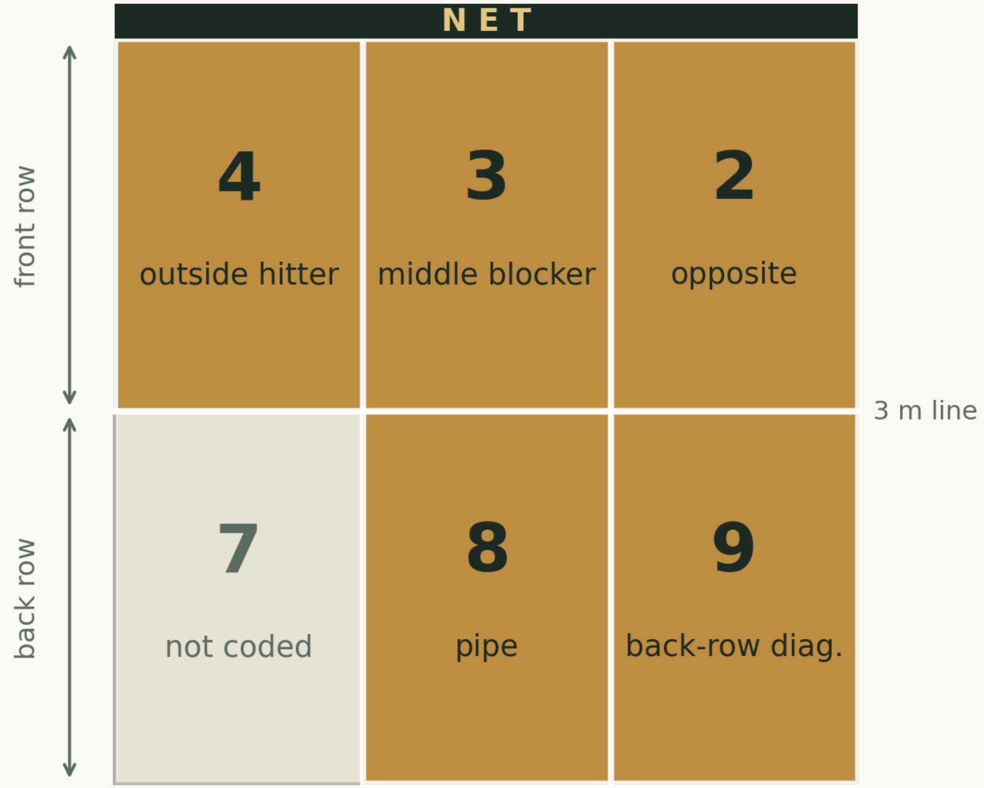
In volleyball, the setter is responsible for organizing the team’s attack. After the reception or defensive action, she usually decides which attacker receives the ball and which type of set to play.

This decision is repeated throughout the match and depends on several factors, including the quality of the first contact, the available attackers and the game situation.

For this project, the setter is therefore particularly interesting: her choices provide an observable sequence of repeated decisions between alternative attackers.



Where an attack comes from: court zones



Data Volley records the zone from which each attack starts. Zones 4, 3 and 2 are front-row attack zones, close to the net.
Zones 8 and 9 represent back-row attacks from behind the 3-metre line.

Different zones are associated with different attacking roles and options.

WHY IT MATTERS

The setter's options depend partly on the players' position on court. Front-row and back-row players have different attacking possibilities, so court position may influence which attacker receives the ball.

RESEARCH QUESTIONS

Real hot hand vs Believed hot hand

RQ1

Real hot hand

Does an attacker's probability of success depend on the outcome of her previous attack?

Previous attack outcome → next attack outcome

This tests whether recent success is associated with subsequent performance

RQ2

Believed hot hand

Does the setter’s next choice depend on the outcome of the attacker’s previous attack?

Previous attack outcome → setter’s next choice

This tests whether the setter behaves as if recent success matters

Does competitive pressure change attacking behaviour?

RQ3

Competitive pressure

Do attackers perform and behave differently during critical phases of a set?

- **Performance:** does the probability of an attack error change under pressure?

- **Attack choice:** does the distribution of attacking tempo change under pressure?

Pressure will be operationalized using the score and the phase of the set.

↓ Choking

Performance deteriorates under pressure

↑ Clutch

Performance improves under pressure

2025-2026 season of Data Volley scouting

63

Data Volley matches scouted

2

Serie A2 teams

2025-26

Same season

The analysis focuses on attacks and the information needed to reconstruct previous outcomes, subsequent choices and score conditions

The two teams under study

Valsabbina Millenium Brescia

Setter: Vittoria Prandi

36 matches

CDA Volley Talmassons FVG

Setter: Francesca Scola

29 matches

Two head-to-head matches are present in both team collections, resulting in 63 unique matches.

Match data were collected from professional Data Volley scouting files (.dvw), which record each technical action in chronological order.

From Datavolley actions to behavioural variables

Each .dvw file contains the chronological sequence of actions performed during a match

1 **Player & team**

Attacker identity → linking consecutive attacks

2 **Skill & evaluation**

Attack, #, = → success / error

3 **Match & set**

Match ID, set 3 → sequence reconstruction

4 **Attack combination**

Quick, medium, high → attacking tempo

5 **Running score**

22-22 → competitive pressure

The chronological structure allows consecutive attacks and setting decisions to be reconstructed within each match

ANALYTICAL APPROACH

Methodology: building the analytical samples

RQ1 – Performance

Does previous success predict subsequent success?

Previous attack by the same player?

Current attack by the same player?

OUTCOME: current attack success

PREDICTOR: previous attack success

RQ2 – Setter choice

Does previous success influence who is chosen next?

Current team attack?

Next attacking attempt by the team?

OUTCOME: same attacker chosen again

PREDICTOR: previous attack success

RQ3 – Competitive pressure

Does pressure change attacking behaviour?

Attack in a statistically critical phase?

Attack in a non-critical phase?

OUTCOME: error rate / attacking tempo

PREDICTOR: critical-phase flag

Methodology: mixed-effects model

STEP 1

Collect 63 Data Volley scouting files

→

STEP 2

Parse & clean: quality checks

→

STEP 3

Build RQ1/RQ2/RQ3 analytical samples

→

STEP 4

Mixed-effects logistic model

→

STEP 5

Interpret & compare results

Attacks are not independent observations.

The same players appear repeatedly, and attacks are also grouped within matches.

Players differ in their baseline performance, and matches differ in their competitive context.

A simple comparison of raw proportions would risk confusing a genuine effect with the fact that some hitters and some matches are simply different from others

WHY IT'S INTERESTING

Expected contribution

1

Separating performance from decision-making

RQ1 and RQ2 make it possible to distinguish whether successful attacks actually tend to be followed by further success, and whether setters behave as if this were the case. If previous success affects setting decisions without predicting subsequent performance, this would be consistent with a belief in the hot hand that is not supported by the observed performance sequence.

2

A real-world behavioural setting

The analysis uses repeated decisions made during professional matches, where choices have real competitive consequences rather than being produced in an experimental task. Using two teams from the same league and season also makes it possible to examine whether similar patterns appear across different team contexts.

The main contribution is not only asking whether the hot hand exists, but whether behaviour changes because people act as if it exists

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RQ1 : Performance

RQ2 : Decision-making

RQ3 : Pressure